# On India's chip manufacturing dream

Dr Suraj Rengarajan, Head Semiconductors Product Group, Applied Materials India

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India's semiconductor chip manufacturing dream is no longer just etched in strategy papers or hollow government declarations. With talent pipelines being built and supply chains cautiously stitched together, it's finally taking shape. Applied Materials is closely involved in this process, according to Dr. Suraj Rengarajan.

"We are fully aligned with India's ambition to become a global semiconductor hub," says Dr. Suraj Rengarajan, Head of the Semiconductors Product Group at Applied Materials India. "With over 20 years of experience in the country, we're here not just to participate – but to help shape it."

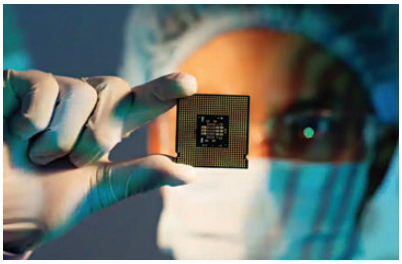
A recent report by ET indicates that the nation's semiconductor market – valued at around US$52 billion in 2024 – is projected to more than double to over US$103 billion by 2030. This surge is fueled by robust initiatives like the US$10 billion India Semiconductor Mission and extended fiscal incentives aimed at fostering local manufacturing and reducing import dependency.

Applied Materials is one of those few names globally that matters in the building blocks of chips, in case you don't know them. Think precision machinery for nanoscale layering and etching, backed by decades of materials science, systems engineering, and process tech. Now they've cast their lot firmly with India.

## India's chip challenges

Because let's face it, building semiconductor manufacturing capacity in India was never going to be easy. It's a generational moonshot. What sets Applied's approach apart, according to Dr. Rengarajan, is their integrated play that brings together infrastructure, innovation, supply chains, and most critically – talent.

In 2023, they launched the Applied Materials Innovation Center for Semiconductor Manufacturing. But that's not just a facility with fancy acronyms – it's a signal, he says. "The center is designed to accelerate the development and commercialization of semiconductor equipment technologies, while building a strong local supply chain," Dr. Rengarajan explains.



And then there's the India Validation Center, or IVC. Located in Bengaluru, it's the first private industry facility in India capable of processing 300-mm wafers. Let that sink in. It's a cornerstone move. Because without the ability to simulate, test, and validate chip-making tools, you're not really building for scale – you're just trialing in theory.

Applied's investments are deliberate, and deeply layered. The IVC supports not just Applied's global roadmap, but for creating a sandbox for Indian fabs, startups, and researchers to plug into.

## Creating Indian talent pool is very important

But chips don't make themselves, and machines certainly don't run themselves. At the heart of this entire movement is talent. And that's where Applied Materials' India team is doubling down in ways that are quietly powerful with deep impact.

"Talent development is a crucial component," says Dr. Rengarajan. "That's why we've partnered with IIT Bombay, IISc, IIT Patna, and IIT Ropar. We want to offer students hands-on experience and direct exposure to industry leaders."

The company's 18-year partnership with IIT Bombay has yielded not just research breakthroughs but actual labs on campus – NanoElectronics and Chemistry & Materials Sciences – and the first-of-its-kind semiconductor certification course. And with the ASCENT program (Applied Semiconductor Collaboration in Engineering and Technology), they're building bridges between faculty, students, and working professionals to take academic curiosity to real-world readiness. d

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